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Prospective Validation of Circulating Tumor DNA Measurable Residual Disease After First-Line Therapy in Large B-Cell Lymphoma

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ABSTRACT

PURPOSE End-of-treatment (EOT) response evaluation by positron emission tomography (PET) remains suboptimal in patients with large B-cell lymphoma (LBCL), because of its limited positive predictive value (PPV). Circulating tumor DNA (ctDNA)—based measurable residual disease (MRD) detection offers a minimally invasive approach and may improve prognostication. We prospectively evaluated EOT MRD using phased variant enrichment and detection sequencing (PhasED-Seq) in patients with first-line LBCL.

METHODS Patients were enrolled in the HOVON-902 prospective cohort and received curative-intent first-line treatment. Phased variants (PVs) were identified and tracked using tumor biopsies or pretreatment plasma. The prognostic significance of EOT ctDNA-MRD status in progression-free survival (PFS) and overall survival (OS) was compared with that of the International Prognostic Index (IPI) and EOT PET-computed tomography (CT).

RESULTS PV identification was successful in 134 of 136 (99%) using either tissue or plasma. At EOT, 83% of patients were MRD-negative and 17% of patients were MRD-positive. MRD positivity was strongly associated with inferior outcomes: the 3-year PFS was 17% in MRD-positive versus 85% in MRD-negative patients (hazard ratio [HR], 9.8 [95% CI, 5.1 to 19]; $P = 9.63 \times 10^{-12}$), and the OS was 43% versus 92%, respectively (HR, 7.7 [95% CI, 3.4 to 17.4]; $P = 1.27 \times 10^{-6}$). In multivariate analysis, MRD was an independent prognostic factor when controlling for IPI and EOT PET-CT. MRD positivity had a higher PPV for 2-year PFS than positive PET (68% *v* 56%, $P \leq .001$), whereas negative predictive value was similar between negative MRD and PET (89% *v* 88%, $P = .71$). MRD positivity was associated with a significantly higher relapse risk within both complete metabolic response (CMR) and non-CMR subgroups.

CONCLUSION This study validates ultrasensitive ctDNA-MRD detection using PhasED-Seq in a uniformly treated, prospective real-world LBCL cohort. These findings support further evaluation of MRD integration into clinical response assessment.

ACCOMPANYING CONTENT

Data Sharing Statement

Data Supplement

Protocol

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INTRODUCTION

For more than a decade, end-of-treatment (EOT) remission assessment in large B-cell lymphoma (LBCL) has relied on ^{18}F-fluorodeoxyglucose positron emission tomography-computed tomography (^{18}F-FDG PET-CT; hereafter, PET-CT), interpreted using the Deauville five-point scale.[1] While PET-CT provides a high negative predictive value (NPV), its positive predictive value (PPV) is limited because of nonspecific FDG uptake in inflammatory or reactive tissues. This limitation often necessitates additional follow-up imaging or biopsies to confirm residual disease, increasing radiation exposure, procedural risk, and patient anxiety.[2-5] Moreover, interobserver variability and the limited accessibility of some lesions for biopsy further complicate clinical decision making.[2] These challenges highlight the need for more accurate and minimally invasive biomarkers to guide

CONTEXT

Key Objective

Can the prognostic value of ultrasensitive circulating tumor DNA measurable residual disease (ctDNA-MRD) assessment at end of treatment (EOT) be validated in a real-world cohort of patients with uniformly treated newly diagnosed large B-cell lymphoma (LBCL)?

Knowledge Generated

The prognostic value of EOT ctDNA-MRD was validated in a uniformly treated first-line LBCL cohort. ctDNA-MRD demonstrated robust independent prognostic value complementary to EOT positron emission tomography-computed tomography (PET-CT) and superior to baseline International Prognostic Index. ctDNA-MRD assessment identified high-risk patients within both EOT PET-negative and PET-positive subgroups.

Relevance *(J.W. Friedberg)*

These data extend other recently published observations demonstrating the value of MRD assessment in identifying patients with diffuse LBCL who have a high risk of relapse, and support ongoing and planned studies of interventions in patients with detectable MRD to abrogate this risk.*

*Relevance section written by *JCO* Editor-in-Chief Jonathan W. Friedberg, MD.

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remission assessment and inform consolidation therapy decisions at the end of therapy.

Circulating tumor DNA (ctDNA)–based measurable residual disease (MRD) detection has emerged as a promising alternative, enabling detection of molecular residual disease with high sensitivity.[6-8] Among these, phased variant enrichment and detection sequencing (PhasED-Seq) has demonstrated superior sensitivity for ctDNA-MRD detection in aggressive LBCLs because of the lower error profile of phased variants (PVs).[8] In a recent study, PhasED-Seq demonstrated a background error rate below one part per 10 million and a validated assay demonstrated a 95% limit of detection of 0.7 parts per million.[9] Previous studies have associated ctDNA-MRD positivity at EOT with poor survival outcomes; however, these studies were limited by small sample sizes, heterogeneous treatment regimens, and single-center cohorts.[8,10,11] The National Comprehensive Cancer Network (NNCN) Guidelines: B-Cell Lymphomas (v3.2025) note that for patients with positive EOT PET-CT findings, repeat biopsy should be considered; when biopsy is not feasible, ctDNA-based MRD testing may be considered (category 2B).[12] Recently, Roschewski et al[13] demonstrated in this journal the prognostic value of EOT ctDNA-MRD in a pooled analysis of clinical trials where patients received experimental therapies. To enable broad clinical adoption, however, ctDNA-MRD–based response assessment requires validation in a uniformly treated LBCL cohort of real-world patients, which reflects routine clinical practice.

In this study, we evaluated the prognostic value of PhasED-Seq–based ctDNA-MRD at EOT in a national cohort of patients with newly diagnosed LBCL uniformly treated with curative-intent first-line immunochemotherapy. We examined the clinical utility of EOT ctDNA-MRD in the context of current standard response assessment.

METHODS

Patients

We analyzed eligible tumor and plasma samples from eligible patients enrolled in the HOVON-902 study (ClinicalTrials.gov identifier: NCT04139252), a multicenter, prospective national real-world cohort designed to validate liquid biopsy biomarkers in newly diagnosed LBCL, as defined by the 2016 WHO Classification.[14] All patients were 18 years and older with previously untreated LBCL and received curative-intent, anthracycline-based immunochemotherapy containing rituximab. The study was approved by the Institutional Review Board of Amsterdam UMC (NL70323.029.19) and conducted in accordance with the Declaration of Helsinki. Patients were enrolled across 46 academic and community centers in the Netherlands between March 2020 and July 2021, and all provided written informed consent. Details on data management are provided in the Data Supplement (online only). The Full Protocol (online only) of HOVON-902 is provided in the Data Supplement.

HOVON-902 was established as a prospective, noninterventional, national study for biomarker discovery in LBCL. The present work represents a subsequent validation study of EOT ctDNA-MRD using PhasED-Seq, conducted according to a prespecified statistical analysis plan (Data Supplement). Additional eligibility criteria and data

management details are provided in the Data Supplement. Clinical annotations of the patients included in analysis are provided in the Data Supplement.

EOT MRD Samples

First-line treatment consisted of rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP) or dose-adjusted rituximab, etoposide, prednisone, vincristine, cyclophosphamide, and doxorubicin (R-EPOCH) at physician discretion. The EOT plasma sample was defined as the first sample collected after completion of chemoimmunotherapy; additional rituximab cycles were permitted, reflecting real-world practice.

Inclusion criteria required patients to have newly diagnosed diffuse LBCL (DLBCL) or high-grade B-cell lymphoma (HGBL), with a baseline sample consisting of formalin-fixed paraffin-embedded (FFPE) tumor tissue and/or pretreatment plasma. In addition, patients were required to have an EOT plasma sample collected within 105 days of day 1 of their final immunochemotherapy cycle, and the EOT plasma draw and the PET-CT had to be performed within 42 days of each other. Exclusion criteria included screening failure or withdrawal of informed consent, histology other than DLBCL/HGBL, and limited-stage (I-II) disease treated with abbreviated R-CHOP plus radiotherapy. Patients were also excluded if they enrolled in a subsequent clinical protocol testing immune checkpoint inhibitor consolidation after EOT sampling (ie, the HOVON-151 or HOVON-152 clinical trials) or if they had primary refractory disease necessitating transition to second-line therapy before EOT evaluation.

EOT Clinical Response Evaluation

EOT PET-CT scans were interpreted locally by nuclear medicine radiologists according to the 2014 Lugano criteria and Deauville five-point scale; scores 1-3 were considered complete metabolic response (CMR), and scores 4-5 were considered non-CMR. Disease progression or relapse during follow-up was assessed per the treating hematologist according to standard of care, with confirmation by PET-CT and/or biopsy where indicated; in cases of rapid clinical progression without imaging, events were accepted if documented in the medical record. Patients achieving CMR were followed at least 24 months according to Dutch national guidelines; only one patient was censored earlier because of emigration. Patients without an event were censored at last follow-up.

EOT MRD Analysis

Patient-specific PVs were identified from FFPE tumor tissue or pretreatment plasma and tracked in EOT plasma using PhasED-Seq, as previously described.[8,9] PV identification was successful if ≥1 PV was identified in the baseline sample and available for monitoring. MRD positivity was defined as detection of one or more PV-containing molecule above a statistically determined threshold corresponding to a false-positive rate of <1%.[9] This threshold was also used to determine the sensitivity for disease detection in each sample. Full methodological details on specimen processing, library preparation, and bioinformatic pipelines are available in the Data Supplement.

Statistical Analysis

Progression-free survival (PFS) and overall survival (OS) were estimated using the Kaplan-Meier method and compared with the log-rank test. Hazard ratios (HRs) with 95% confidence intervals (CIs) were calculated using Cox proportional hazards regression. Candidate predictors for multivariable models included baseline International Prognostic Index (IPI), EOT PET-CT response (CMR *v* non-CMR), and EOT ctDNA-MRD status (negative *v* positive). For relapse-focused analyses, time to progression was defined as the interval from treatment initiation to lymphoma-related progression or relapse, censoring non–lymphoma-related events. Censored patients were considered nonevents in binary classification analyses.

To compare the predictive performance of baseline IPI, EOT PET-CT, and EOT ctDNA-MRD in 2-year PFS, we applied predefined dichotomizations: IPI ≥ 3 (*v* 0-2), PET-CT non-CMR (*v* CMR), and ctDNA-MRD–positive (*v* ctDNA-MRD–negative). Patients with events at any time were included; those event-free but censored before 24 months were excluded from fixed-horizon performance estimates. For each test, sensitivity, specificity, PPV, and NPV at 2 years were calculated using Wilson 95% CIs.

When comparing EOT PET-CT and EOT ctDNA-MRD, pairwise McNemar's tests with Holm-adjusted *P* values were reported. PPV and NPV were compared using generalized estimating equation logistic regression (exchangeable correlation structure, robust standard errors), with a global Wald test for the test factor and Holm-adjusted pairwise contrasts (reported as odds ratios with 95% CIs). Absolute differences in PPV and NPV between tests were estimated using nonparametric patient-level bootstrap resampling (10,000 replicates; percentile 95% CIs).

Discrimination was quantified using Harrell's concordance index from univariable Cox models, with 95% CIs estimated by bootstrap. Associations between MRD status and clinicopathologic features were assessed using Fisher's exact test. Statistical significance was defined as a two-sided $P < .05$. Analyses were conducted using R version 4.4.3 (R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

Patient Characteristics and Response Evaluation

A total of 250 patients were enrolled in HOVON-902. After exclusions, 136 patients were eligible for this study. A

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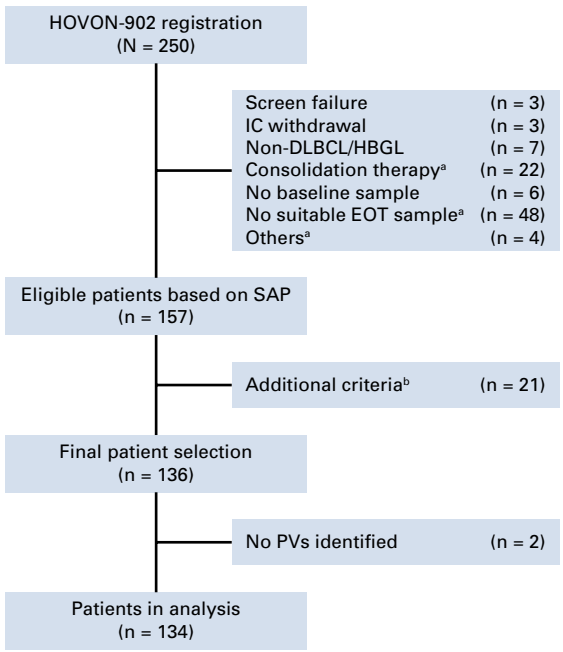

B

Characteristic	No. (n = 134)	Percentage
Median age, years	69 (19-88)	
Sex		
Male	84	63
Female	50	37
Diagnosis		
DLBCL	128	96
HGBL-DH/TH	6	4
Ann Arbor stage		
I-II	28	21
III-IV	106	79
IPI		
0-1	28	21
2	45	34
3	33	25
4-5	28	21
1L therapy regimen		
R-CHOP	128	96
DA-EPOCH-R	6	4
Cycles of treatment		
6 cycles	132	99

FIG 1. Patient inclusion and baseline characteristics. (A) Diagram of patient enrollment and sample inclusion for EOT ctDNA-MRD analysis. The HOVON-902 study registered 250 patients, of whom 157 were eligible for this study based on the prespecified SAP (Data Supplement). [a]Among the 22 patients who received consolidation therapy, 15 received an immune checkpoint inhibitor and seven received radiotherapy. Of the 48 patients without an evaluable EOT sample, 15 died before EOT from complications (including infections such as COVID-19 disease), 22 discontinued protocol-mandated blood sampling while continuing therapy, and 13 had plasma collected on C6D1 or earlier. Among the four additional exclusions, one received long-term rituximab for another condition, one underwent tumor debulking before treatment, one had a concurrent solid tumor, and one did not receive standard frontline therapy. [b]Additional eligibility criteria are provided in the Data Supplement. Seven patients with primary refractory disease were excluded. Fourteen patients were excluded because of EOT plasma collection >105 days after C6D1 or an interval of >42 days between EOT PET and MRD sampling. (B) Baseline characteristics of the patients in analysis. C6D1, cycle 6 day 1; ctDNA, circulating tumor DNA; DLBCL, diffuse large B-cell lymphoma; EOT, end-of-treatment; HGBL, high grade B-cell lymphoma; HGBL-DH/TH, high grade B-cell lymphona - double hit/triple hit; IC, informed consent; IPI, International Prognostic Index; MRD, measurable residual disease; PET, positron emission tomography; PV, phased variant; SAP, statistical analysis plan.

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detailed breakdown of exclusions is provided in the CONSORT diagram (Fig 1A). PVs were successfully identified in 134 of 136 eligible patients (99%). Figure 1B shows the baseline patient characteristics of the patients included in this analysis. Age ranged from 19 to 88 years, with a median of 69 years, and most patients were male (63% male *v* 37% female). The majority had de novo DLBCL (96%), and 79% of patients presented with advanced-stage disease (Ann Arbor stage III/IV). The distribution of IPI risk groups was 21%, 34%, 25%, and 21% for low, low-intermediate, high-intermediate, and high IPI, respectively. All patients received curative-intent first-line anthracycline-based immunochemotherapy containing rituximab: 96% received R-CHOP and 4% received DA-EPOCH-R, with 99% of all patients completing all six cycles.

At EOT, 131 patients were evaluated by PET-CT, and one by CT, and two patients did not have EOT imaging. Of those evaluated by PET-CT, 103 (79%) were in CMR, whereas 28 (21%) patients had non-CMR. EOT ctDNA-MRD status was successfully determined in all 134 patients. Of the 134 patients included in analysis, 99% had a sample-level sensitivity below 0.01% (1 in 1e4), 73% had sensitivity below 0.001% (1 in 1e5), and 3% had sensitivity below 0.0001% (1 in 1e6). Sample-level sensitivity is directly related to the input used, and the median plasma volume was 4.6 mL, whereas the median DNA input was 19.8 ng. The distribution of sample-level sensitivities and their relationship with the number of assessed PVs in each case are provided in the Data Supplement (Supplementary Notes and Fig S1). The median follow-up time was 33.2 months (range, 3.4-53.9 months); the 3-year PFS and OS were 74% (95% CI, 66 to 82) and 84% (95% CI, 77 to 91), respectively (Data Supplement, Fig S2).

EOT ctDNA-MRD Is Prognostic of PFS and OS

At EOT, 111 of 134 patients (83%) were ctDNA-MRD–negative, whereas 23 of 134 (17%) were ctDNA-MRD–positive. MRD positivity was significantly associated with advanced disease stage (Ann Arbor stage III/IV; $P = .045$;

Data Supplement, Fig S3A), but not significantly associated with higher IPI risk groups (P = .061; Data Supplement, Fig S3B). In the entire cohort, irrespective of the number of treatment cycles or imaging-based response, ctDNA-MRD status in plasma collected after 1L immunochemotherapy was strongly prognostic for long-term outcomes. MRD-positive patients had significantly inferior 36-month PFS compared with MRD-negative patients (17% *v* 85%; HR, 9.8 [95% CI, 5.1 to 19]; log-rank $P = 9.63 \times 10^{-12}$; Fig 2A). A similar pattern was observed for OS, with 3-year OS rates of 43% in MRD-positive patients versus 92% in MRD-negative patients (HR, 7.7 [95% CI, 3.4 to 17.4]; log-rank $P = 1.27 \times 10^{-6}$; Fig 2B).

Prognostic Value of EOT ctDNA-MRD Is Independent of IPI and PET-CT

Next, we benchmarked the prognostic value of EOT ctDNA-MRD against established clinical risk stratification tools, including the baseline IPI and the EOT PET-CT scan, in the subset of patients with all three data available (n = 131). In this subgroup, IPI was not prognostic for PFS (log-rank P = .46) and OS (log-rank P = .98; Fig 3A and Data Supplement, Fig S4A). The EOT PET-CT scan was prognostic for both PFS (HR, 6.2 [95% CI, 3.2 to 11.9]; log-rank $P = 6 \times 10^{-8}$) and OS (HR, 5.8 [95% CI, 2.5 to 13.4]; log-rank $P = 4.45 \times 10^{-5}$; Fig 3B and Data Supplement, Fig S4B). EOT ctDNA-MRD showed the strongest prognostic value for PFS (HR, 9.3 [95% CI, 4.8 to 18.1]; log-rank $P = 5.19 \times 10^{-11}$) and OS (HR, 7 [95% CI, 3 to 16.2]; log-rank $P = 5.68 \times 10^{-6}$; Fig 3C and Data Supplement, Fig S4C). To further compare these risk stratification tools, we performed univariate and multivariate Cox regression analyses. Univariate analysis confirmed the prognostic significance of EOT PET-CT and MRD in PFS and OS (Fig 3D and Data Supplement, Fig S4D), and EOT MRD remained independently prognostic in multivariate models for both PFS and OS (Fig 3E and Data Supplement, Fig S4E). The Data Supplement (Table S1) summarizes the univariate and multivariate Cox regression analyses of IPI, PET-CT, and ctDNA-MRD.

Sensitivity, specificity, PPV, and NPV of EOT PET-CT and ctDNA-MRD for predicting 2-year PFS were compared (Fig 3F). MRD positivity had a higher PPV for 2-year PFS than positive PET (68% *v* 56%, $P \leq .001$), whereas NPV was similar between negative MRD and PET (89% *v* 88%, P = .71).

ctDNA-MRD Identifies High-Risk Patients in Both EOT CMR and Non-CMR Patients

We assessed the prognostic value of ctDNA-MRD within both CMR and non-CMR patients as defined by EOT PET-CT. Among 103 patients who achieved CMR, nine (9%) were MRD-positive and 94 (91%) were MRD-negative. Within this subgroup, MRD positivity was associated with significantly worse 3-year PFS compared with MRD-negative patients (44% *v* 88%; HR, 5.4 [95% CI, 1.9 to 15.1]; log-rank $P = 3.69 \times 10^{-4}$), whereas no significance for OS was observed (88% *v* 92%; HR, 1.1 [95% CI, 0.1 to 8.5]; log-rank P = .95; Fig 4A and Data Supplement, Fig S5A). In the non-CMR group (n = 28), 13 (46%) were MRD-positive and 15 (54%) were MRD-negative. MRD-positive patients had a 0% 3-year PFS, compared with 66% in MRD-negative patients (HR, 6.3 [95% CI, 2.1 to 18.2]; log-rank $P = 1.85 \times 10^{-4}$), and a similar pattern was observed for OS (17% *v* 86%; HR, 9.1 [95% CI, 2.1 to 45.1]; log-rank $P = 4.66 \times 10^{-4}$; Fig 4B and Data Supplement, Fig S5B).

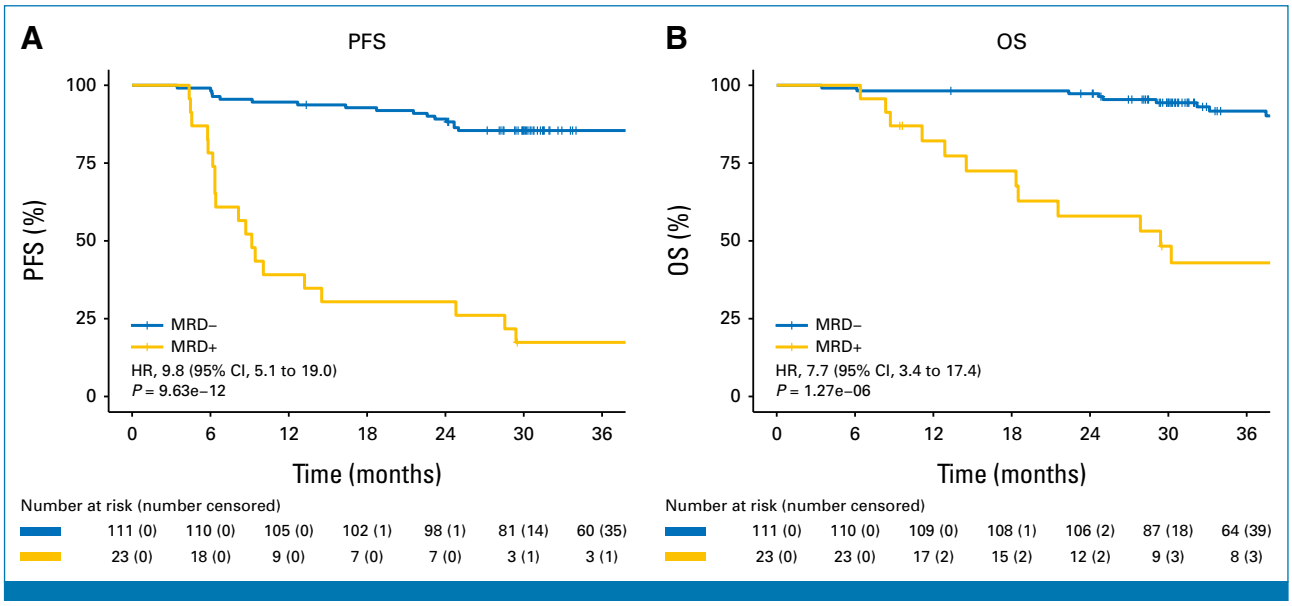

FIG 2. EOT ctDNA-MRD is prognostic for PFS and OS. (A) Kaplan-Meier estimates of PFS stratified by EOT ctDNA-MRD status. (B) Kaplan-Meier estimates of OS stratified by ctDNA-MRD status. ctDNA, circulating tumor DNA; EOT, end-of-treatment; HR, hazard ratio; MRD, measurable residual disease; OS, overall survival; PFS, progression-free survival.

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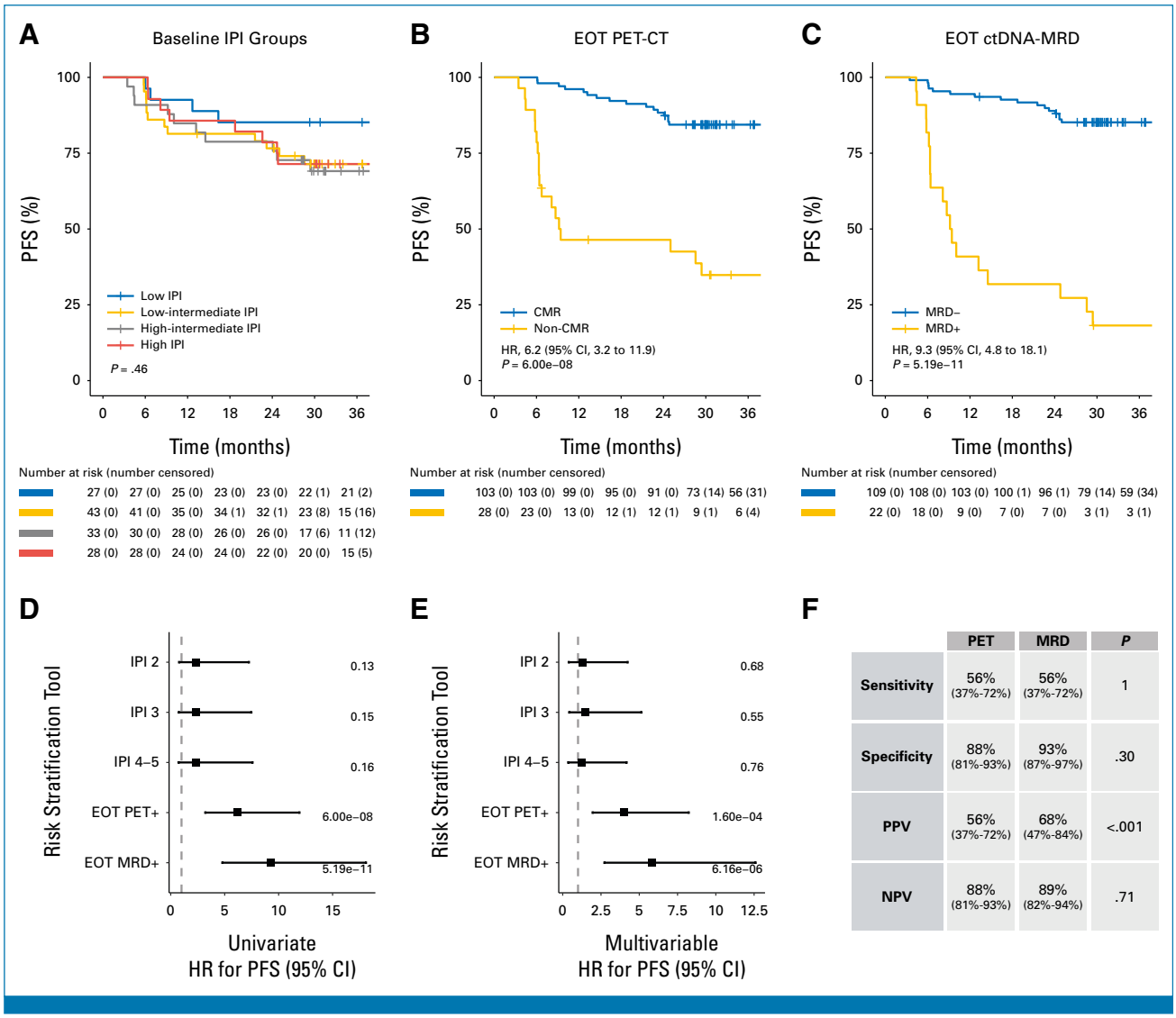

	PET	MRD	P
Sensitivity	56% (37%-72%)	56% (37%-72%)	1
Specificity	88% (81%-93%)	93% (87%-97%)	.30
PPV	56% (37%-72%)	68% (47%-84%)	<.001
NPV	88% (81%-93%)	89% (82%-94%)	.71

FIG 3. Prognostic performance of baseline IPI, EOT PET-CT, and ctDNA-MRD. (A) PFS stratified by baseline IPI risk groups. (B) PFS stratified by first EOT PET-CT. (C) PFS stratified by EOT ctDNA-MRD. (D) Univariate Cox proportional hazards models for PFS. (E) Multivariate Cox proportional hazards models for PFS. (F) Performance of baseline EOT PET-CT and EOT ctDNA-MRD in predicting 2-year PFS events. *P* values for sensitivity and specificity were obtained using McNemar's test with Holm adjustment. *P* values for PPV and NPV were obtained from GEE logistic regression with Holm-adjusted pairwise contrasts. Patients with a <2-year PFS follow-up were excluded. CMR, complete metabolic response; ctDNA, circulating tumor DNA; EOT, end-of-treatment; GEE, generalized estimating equation; HR, hazard ratio; IPI, International Prognostic Index; MRD, measurable residual disease; NPV, negative predictive value; PET-CT, positron emission tomography-computed tomography; PFS, progression-free survival; PPV, positive predictive value.

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EOT MRD Predicts Early Relapses

Finally, we assessed whether ctDNA-MRD positivity at EOT predicted disease relapse and whether it was associated with early or late recurrence. Among 27 lymphoma-related relapses, 16 occurred within 12 months (early) and 11 occurred after 12 months (late) since the start of treatment. EOT MRD was positive in 75% (12 of 16) of the patients with early relapse and 27% (3 of 11) of the patients with late relapse. EOT ctDNA-MRD detected a higher proportion of early relapses compared with late relapses (Fisher's exact test, $P = .022$; Fig 4C).

DISCUSSION

Ultrasensitive ctDNA-based MRD at EOT is a powerful and independent predictor of clinical outcomes in first-line LBCL. In this prospective, real-world validation study, we demonstrate that PhasED-Seq–based MRD testing stratifies patients by relapse risk with greater discriminatory power than baseline IPI or EOT PET-CT. Among patients with positive MRD results at EOT, relapse was common and outcomes were poor, whereas those who tested negative had durable remissions. Specifically, we demonstrate that MRD-negative patients with non-CMR

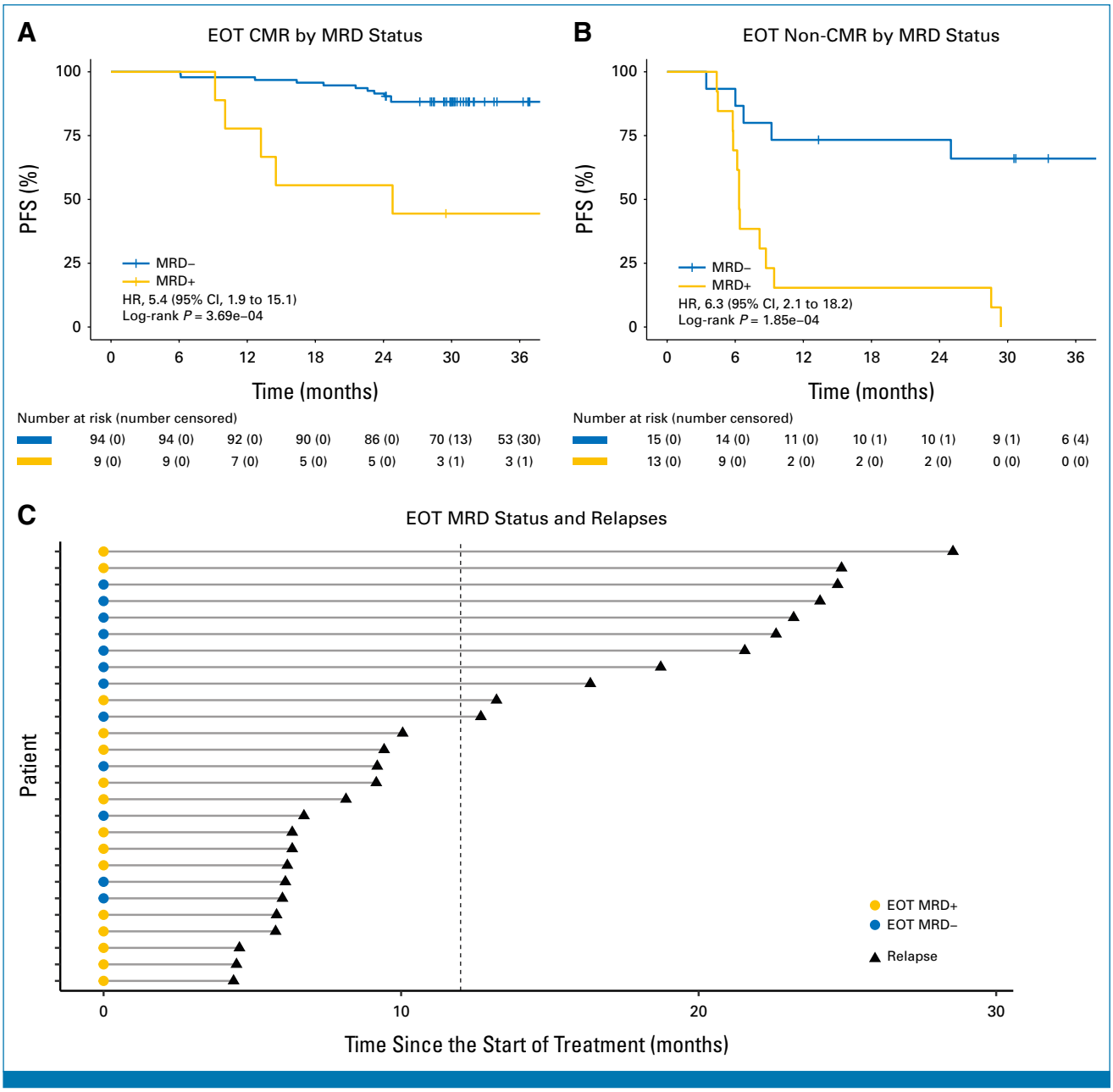

FIG 4. Prognostic value of EOT ctDNA-MRD within PET-defined subgroups. (A) PFS stratified by MRD status in patients with CMR. (B) PFS stratified by MRD status in patients without CMR. (C) Swimmer plot showing EOT MRD status and timing of relapse in individual patients. The vertical dashed line represents 12 months since the end of first-line treatment. CMR, complete metabolic response; ctDNA, circulating tumor DNA; EOT, end-of-treatment; HR, hazard ratio; MRD, measurable residual disease; PET, positron emission tomography; PFS, progression-free survival.

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PET-CT had significantly improved PFS compared with MRD-positive patients, suggesting the clinical utility of using ctDNA to adjudicate potentially false-positive metabolic activity.

To support broad clinical utility, it is important that the technology is applicable to the wide range of patients with LBCL. Reassuringly, PVs were successfully identified in 99% of patients using either baseline plasma or diagnostic FFPE tissue. This flexibility in the baseline sample for PV identification enhances assay applicability, particularly when either tissue or pretreatment plasma samples might be unavailable. Moreover, we demonstrated the feasibility of centralized ctDNA analysis using plasma samples collected in academic and community centers nationwide, underscoring the logistical scalability of the assay in clinical practice.

EOT ultrasensitive ctDNA-MRD positivity was strongly associated with inferior PFS and OS. In direct comparisons, ctDNA-MRD was independently prognostic for PFS and complementary to both IPI and PET-CT. These findings support the prognostic relevance of ctDNA-MRD, as recently

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demonstrated in this journal by Roschewski et al,[13] and build upon these recent findings for two key reasons. First, this analysis was conducted in a uniformly treated cohort of patients with first-line LBCL, in contrast to selected trial populations which are often enriched for more fit individuals. Second, the previous study considered a variety of therapies including investigational agents, where all patients in our cohort received standard curative-intent first-line treatment regimens, confirming the prognostic value of EOT ultrasensitive ctDNA-MRD testing in a real-world setting.

Demonstrating robust assay performance is critical for clinical adoption of ctDNA-MRD. For example, although a study of ctDNA analysis via immunoglobulin sequencing initially showed strong performance in retrospective analyses of samples collected over 2 decades,[7] interim data from a prospective validation study by Kumar et al[15] demonstrated only modest results. More recently, data from the study by Nijland et al[16] showed suboptimal clinical sensitivity of immunoglobulin sequencing in the HOVON-151 cohort (first-line DLBCL patients with IPI ≥ 3 treated with R-CHOP). Together, these observations highlight that rigorous validation in well-defined, contemporary cohorts is paramount for clinical translation of ctDNA-MRD assays.

Subgroup analyses underscore the clinical utility of MRD in current practice. All non-CMR patients with positive MRD relapsed within 3 years, identifying a subset at extremely high risk that warrants further study in clinical trials evaluating treatment intensification strategies. Conversely, MRD-negative patients with non-CMR PET-CT scans had a 3-year PFS of 66%, with two of five PFS events unrelated to lymphoma. These findings suggest that ctDNA-MRD may help distinguish true residual disease from false-positive metabolic activity, potentially avoiding unnecessary biopsies or escalation of therapy in patients with persistently positive PET-CT. This aligns with current NCCN guidelines, which suggest that an ultrasensitive MRD assay to confirm residual disease may be used as an alternative to biopsy when EOT PET-CT is positive and biopsy is not feasible.[12]

Importantly, the immediate utility of treatment based on EOT MRD positivity alone is not yet proven. We thus do not advocate initiating consolidation therapy solely based on an EOT MRD-positive result without prospective studies exploring this approach. While ongoing trials such as ALPHA3 (ClinicalTrials.gov identifier: NCT06500273) and GO43075 (ClinicalTrials.gov identifier: NCT04980222) are testing MRD-directed strategies in LBCL,[17,18] identifying high-risk patients who truly benefit from escalation remains challenging. Other investigative strategies have relied on clinical risk scores (eg, IPI ≥ 3, as in HOVON-151[16]) or metabolic response (eg, interim PET in PETAL[19] or EOT PET in HOVON-152[20]) although these tools have suboptimal PPV. Nevertheless, trials such as ALPHA3 demonstrate the feasibility of performing real-world EOT MRD testing in the clinic with assay turnaround times of under 14 days, allowing clinical decision making on the basis of these results.

EOT ultrasensitive MRD detected most patients who relapsed within 1 year after first-line treatment, compared with 27% among those with late relapses. This reinforces the clinical value of MRD for identifying early relapses in patients who may be candidates for second-line therapies, such as CAR-T-cell therapy. These findings highlight the potential utility of additional repeat post-treatment MRD testing to improve detection of late relapses and to inform long-term surveillance strategies. Improving the sensitivity of tested samples by maximizing the amount of analyzed material (ie, by increasing the amount of plasma input) may lead to improved sensitivity for late relapses. Alternatively, a follow-up testing approach incorporating additional follow-up samples may be desirable to capture late relapses with biologically indolent or slowly emerging disease. Eight MRD-positive patients remained relapse-free until nonlymphoma death or the last follow-up, potentially because of relapses that had not yet occurred, remaining indolent lymphoma (one such patient had a composite lymphoma), or delayed ctDNA-MRD clearance (eg, from long-acting antibodies like rituximab). In addition, MRD kinetics during treatment may provide further insight for response-adapted strategies under investigation, such as de-escalation in patients achieving early molecular clearance. The ongoing SHORTEN-ctDNA study (ClinicalTrials.gov identifier: NCT06693830) will investigate such approaches by integrating interim PET-CT with PhasED-Seq-based MRD assessment.[21]

This study has several limitations. First, EOT PET-CT scans were not centrally reviewed, which may introduce variability in imaging interpretation; however, this also reflects real-world clinical practice.[22] Second, the timing of EOT plasma collection (typically 3-9 weeks post-treatment) could potentially affect ctDNA detectability; however, this broad time window is also reflective of real-world practice. Third, some deaths unrelated to lymphoma were counted as PFS events, which may affect outcome estimates in small subgroups. Fourth, sample-level sensitivity was lower than that in previous reports,[13] likely because of lower plasma volumes and prolonged storage, whereas ongoing trials using larger, freshly processed samples have shown higher sensitivity. Fifth, we included MRD samples collected during additional rituximab administration at some centers after completion of immunochemotherapy. Although it is unclear whether such treatment could have contributed to MRD clearance, we consider this unlikely to affect a substantial proportion of patients. Importantly, this approach reflects real-world practice as PET scans were often performed concurrently with MRD sampling. Finally, because all patients were enrolled between 2020 and 2021, the COVID-19 pandemic contributed to protocol deviations (eg, avoiding additional visits for blood collection to reduce hospital exposure).

In summary, we prospectively validated the prognostic utility of EOT ctDNA-MRD assessment using PhasED-Seq in a national, uniformly treated cohort of patients with newly

diagnosed LBCL. Our results demonstrate that ctDNA-MRD is a robust and independent predictor of long-term outcomes, providing additional prognostic value beyond established clinical tools across key patient subgroups, and support its further evaluation for incorporation into response assessment in patients with LBCL.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Prospective Validation of Circulating Tumor DNA Measurable Residual Disease After First-Line Therapy in Large B-Cell Lymphoma

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